Simulation Guide

for *Investigating Lesion Registration Uncertainty in Dynamic Thermography Using Fiducial Leverage and Residuals* (IEEE ISBI 2027, under review) · **Wei-Che Hung**, Department of Systems and Control Engineering, University of Malta · 30 October 2026 · release v1.0.1-isbi2027. What is simulated, which script produces each figure and table, and how to run them; numbering follows the paper.

1. What the simulation tests

Four fiducial markers are registered frame by frame with an affine model, eq. (1). The paper predicts the registration error at a lesion point of interest (POI) from two quantities that every recording with four or more markers provides:

- the **leverage** h_p of the POI, eq. (2), computed once from the marker and POI coordinates in frame-0;
- the **marker residuals** $R^{(k)}$ and their covariance $\hat{\Sigma}^{(k)}$, eq. (3), pooled over the sequence into Σ_{eff}, eq. (4).

The prediction $\mathrm{RMS}_p = \sqrt{h_p \,\mathrm{tr}(\Sigma_{\text{eff}})}$, eq. (5), extends to the intra-lesion error RMS_L, eq. (6), and to the boundary area disagreement $\mathbb{E}[D_A]$, eq. (7); each is compared with the Monte Carlo value from the same trials.

2. Protocol (paper Sec. 3.4)

Item	Value
Image	640 × 480 pixels
Markers	four; *baseline* configuration, and *expanded* configuration with each marker moved 1.5 times farther from the marker centroid
POIs	65 (a 9 × 7 grid plus two interior points); ray of 100 POIs from the marker centroid beyond the marker convex hull, σ = 1 pixel
Motion per trial	affine: rotation ±5°, translation ±10 pixels, scaling ±2 %
Marker noise	isotropic Gaussian, σ ∈ {0.5, 1, 2, 4} pixels, added to the moved markers
Trials	10 000 per noise level, seed 0 (bootstrap band of Fig. 4: seed 1, 1000 resamples)
Lesion polygons	triangle, circle, rectangle of equal area at the marker centroid (δ = 0); circles at normalised offsets δ ∈ {0.5, 1.0, 1.5}; σ = 1 pixel, repeated at σ = 2 for Sec. 4.2

Every script uses the same trial procedure and draw order: draw the motion, move markers and POIs, add marker noise, fit the affine map back to frame-0, record POI errors and marker residuals. The prediction uses residuals and leverage only; the truth uses the known POI positions.

3. Scripts, outputs and paper objects

Commands run from the repository root. Script names keep the draft numbering (`fig1` → Fig. 2, `fig2` → Fig. 3, `fig3` → Fig. 4). Only `fig1_leverage_map.py` asserts pinned numbers before drawing; the others write their statistics to `results/`.

Script	Produces	Paper object
`src/analysis1/fig1_leverage_map.py`	leverage map, worked frame (seed 1, σ = 1): $h = 1.7806$ at $p_1 = (40, 40)$	Fig. 2
`src/analysis1/fig2_probe_rms.py`	predicted vs simulated RMS for the 65 POIs, both configurations, four noise levels; `results/fig2_probe_rms_stats.json`	Fig. 3, Table 1a
`src/analysis2/fig3_polygon_rms.py`	ray sweep with the 95 % simultaneous band; six lesion polygons with 95 % confidence intervals; `results/fig3_polygon_rms_stats.json`	Fig. 4, Table 1a
`src/analysis3/area_error.py`	predicted and simulated area disagreement D_A for the six polygons; `results/table1_area_error_stats.json`	Table 1b
`src/analysis3/table1_render.py`	the 18 numbers of Table 1 from the three stats files	Table 1

Modules: `src/common/affine.py` (geometry, motion, noise), `src/analysis1/mc_core.py` (trial procedure, Fig. 3). Tests: `python -m unittest discover -s <dir>` for `src/common`, `src/analysis1`, `src/analysis2`, `src/analysis3`.

Run.

```
python src/analysis1/fig1_leverage_map.py
python src/analysis1/fig2_probe_rms.py --seed 0 --trials 10000
python src/analysis2/fig3_polygon_rms.py --seed 0 --trials 10000 --sigma 1
python src/analysis2/fig3_polygon_rms.py --seed 0 --trials 10000 --sigma 2 \
    --out results/fig3_polygon_rms_sigma2.pdf --stats results/fig3_polygon_rms_sigma2_stats
python src/analysis3/area_error.py --seed 0 --trials 10000 --sigma 1
python src/analysis3/table1_render.py
```

Requirements: Python 3.10 or later, NumPy, Matplotlib (SciPy for the analysis3 tests only). Figures go to `paper/figures/`, statistics to `results/`; Table 1 is read from those files at render time.

Code availability and contact. This repository holds the guide; the scripts are released here after the review round. Wei-Che Hung, wei-che.hung@um.edu.mt or wolfmib@gmail.com.